

344.175 VL: Natural Language Processing

Word Representation



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Agenda

- Distributional semantics & word embedding
- Point Mutual Information
- GloVe

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- **Distributional semantics & word embedding**
- Point Mutual Information
- GloVe

Meaning & Semantics

Meaning:

What is meant by a word, text, concept, or action. (Oxford)

The thing one intends to convey especially by language. (Merriam-Webster)

Semantics:

The branch of linguistics and logic concerned with meaning ... *lexical semantics* [is] concerned with the analysis of word meanings and relations between them. (Oxford)

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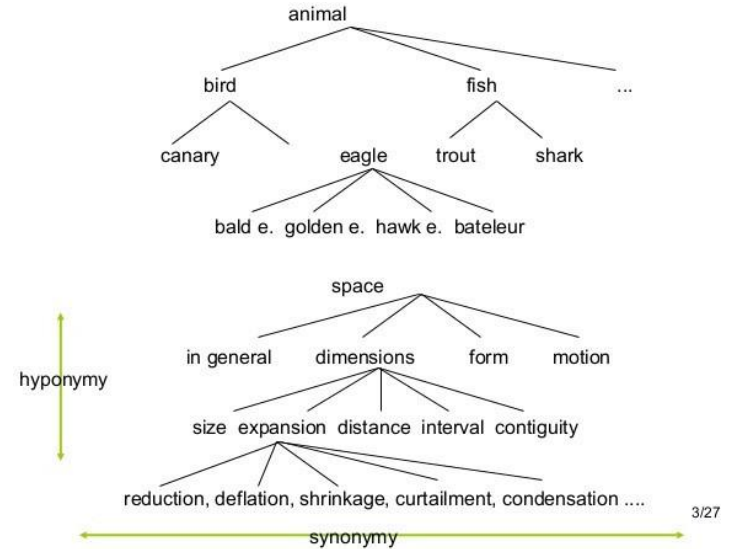


Semantics with knowledge resources



Noun language has 6 senses

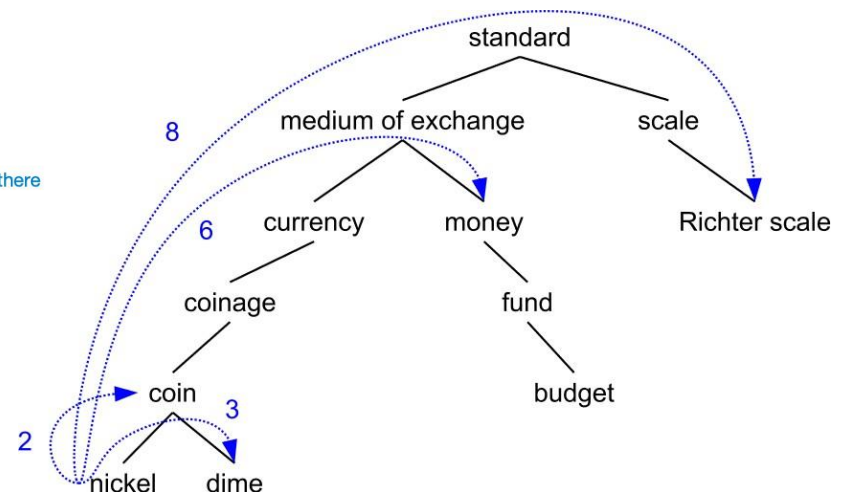
1. **language, linguistic communication** - a systematic means of communicating by the use of sounds or conventional symbols; "he taught foreign languages"; "the language introduced is standard throughout the text"; "the speed with which a program can be executed depends on the language in which it is written"
 --1 is a kind of **communication**
 --1 has particulars:
 dead language; words; **source language**; object language, target language; sign language, signing; artificial language; metalanguage; native language; natural language, tongue; lingua franca, interlanguage, koine; string of words, word string, linguistic string; barrage, outpouring, onslaught; slangue
2. **speech, speech communication, spoken communication, spoken language, language, voice communication, oral communication** - (language) communication by word of mouth; "his speech was garbled"; "he uttered harsh language"; "he recorded the spoken language of the streets"
 --2 is a kind of **auditory communication**
 --2 has particulars:
 words; pronunciation, orthoepy; conversation; discussion, give-and-take, word; saying, expression, locution; non-standard speech; idiolect; monologue; spell, magic spell, charm; dictation; soliloquy, monologue
3. **terminology, nomenclature, language** - a system of words used in a particular discipline; "legal terminology"; "the language of sociology"
 --3 is a kind of **word**
 --3 has particulars: markup language; toponymy, toponomy



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Noun elephant has 2 senses

1. **elephant - five-toed pachyderm**
 --1 is a kind of proboscidean, proboscidian
 --1 is a member of Elephantidae, family Elephantidae
 --1 has particulars:
 rogue elephant; Indian elephant, Elephas maximus; African elephant, Loxodonta africana; mammoth; gomphothere
2. **elephant - the symbol of the Republican Party; introduced in cartoons by Thomas Nast in 1874**
 --2 is a kind of emblem, allegory



Read more here: <https://web.stanford.edu/~jurafsky/slp3/C.pdf>

<https://wordnet.princeton.edu>

<http://wordnet-online.freedicts.com/definition?word=language>

<https://www.slideshare.net/AhmedAbdElwasaa/wordnet-a-database-of-lexical-relations>

Distributional Semantics



“You shall know a word
by the company it
keeps!”

*J. R. Firth, A synopsis of
linguistic theory 1930–1955
(1957)*

Distributional Semantics

“A word’s **meaning** is given by the words that frequently **appear close-by**”

- The **context** of a word w is the set of words appear in the nearby of w (e.g., within a fixed-size window)
 - The words in context are **context-words**
- We use many contexts of w to create a representation of w
 - e.g., the meaning of *banking*

*...government debt problems turning into **banking** crises as happened in 2009...*
*...saying that Europe needs unified **banking** regulation to replace the hodgepodge...*
*...India has just given its **banking** system a shot in the arm...*

One of the most successful ideas of modern statistical NLP!



The Tarahumara people gather every year during Easter week (*semana santa*) and drink large amounts of Tesgüino together while following rituals. According to the anthropologist [Bill Merrill](https://en.wikipedia.org/wiki/Bill_Merrill) of the Smithsonian Institute, the sacred drink chases large souls from the persons who drink it, «and so when people get drunk that's why they act like children [...] because the souls that are controlling their actions are the little souls, like little children».

<https://en.wikipedia.org/wiki/Tesgüino>

Nida[1975]

A word cloud centered around the word "Tesgüino". The words are arranged in a circular pattern around the central term. The words "drink", "beer", and "brew" are colored blue, while the others are black. The words are tilted at various angles.

drink

sacred

beer

Tesgüino

ritual

corn

brew

Mexico

Tarahumara people

dark brown

bottle

malt

brew

lager

Märzen

Bavaria

bar

drink

beer



Tesgüino $\longleftrightarrow$ Märzen



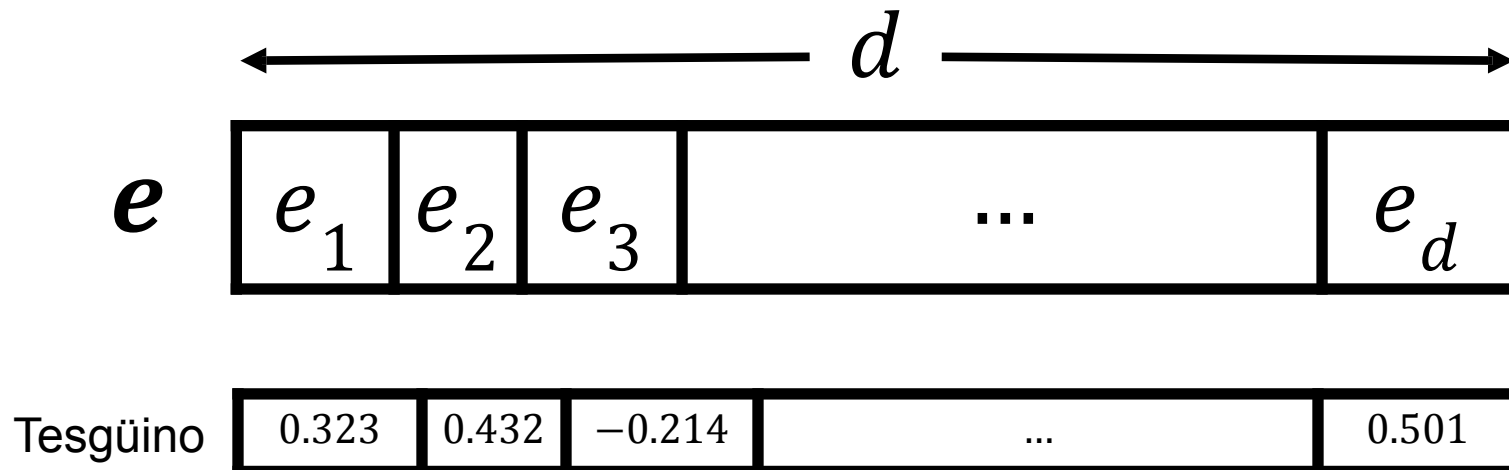
An algorithmic intuition:

Two words are **related (semantically similar)** when they have many **common context-words**

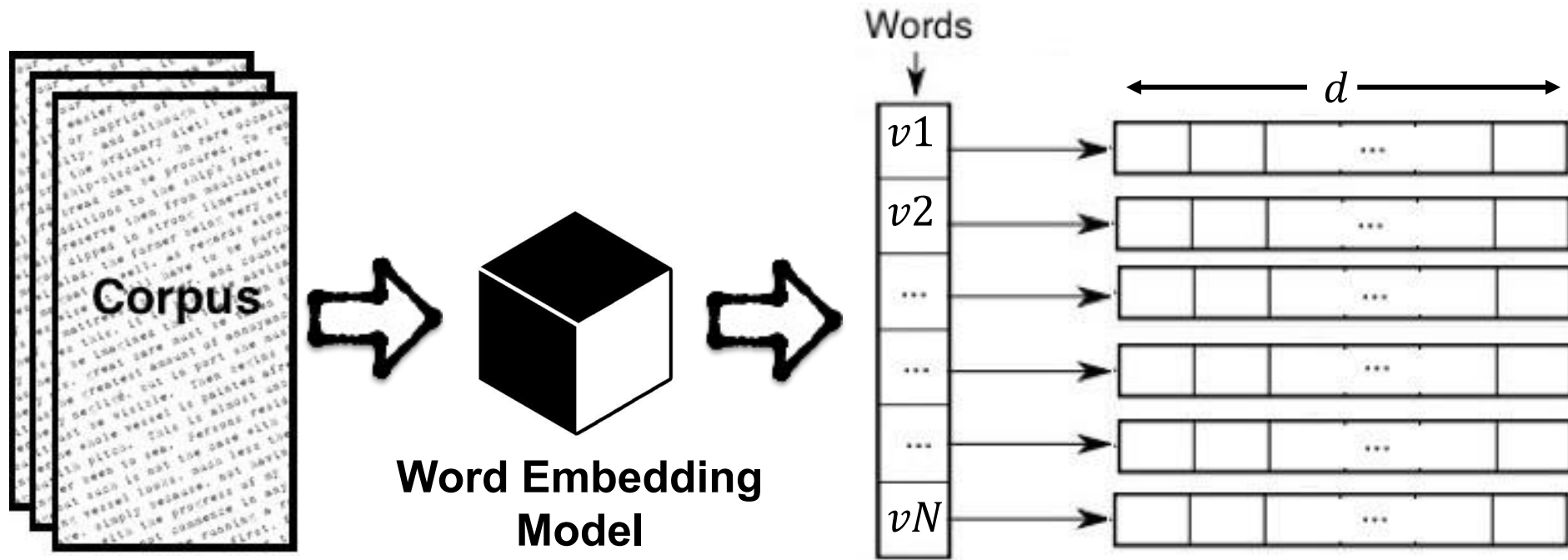
How to represent entities/tokens/information/knowledge?

Distributional Representation

- A word (token) is represented with a **vector of d dimensions**
- Each dimension can be seen as a “**feature**” of the entity (word/token)
- Dimensions and their values are not mutually exclusive
 - Two units can be “active” at the same time
- Each dimension forms a **distribution** over possible values (domain $\mathbb{R}$)
 - Realizing a word vector can be seen as, for each dimension, selecting a value according to the its underlying distribution

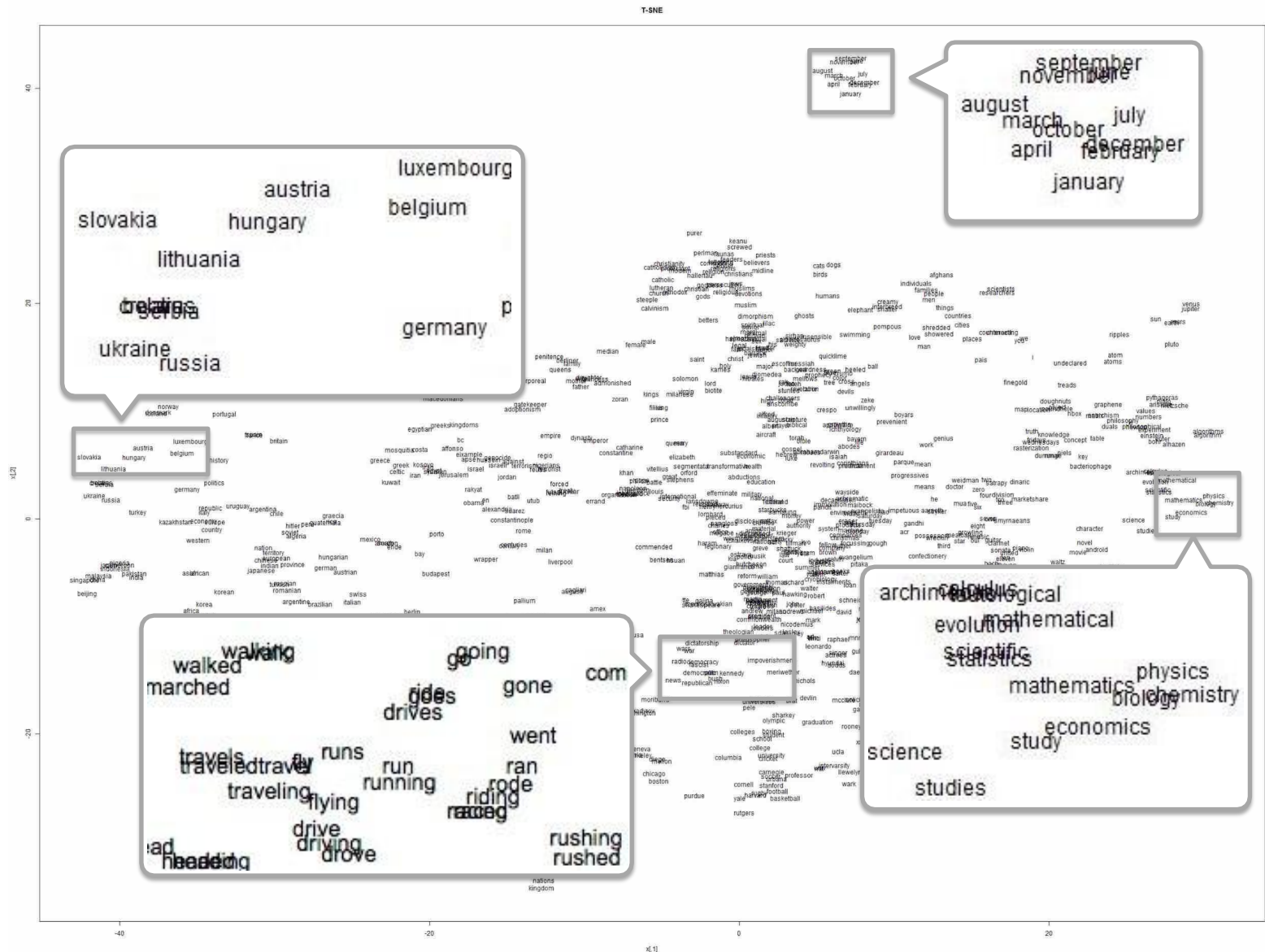


Word Embedding



- We use many contexts of words in the corpus to create vector representations of words
- When vector representations are dense, they are often called **embedding** ... e.g., word embedding

Word Embedding – Visualization



Word embeddings projected to a two-dimensional space

Word Embedding – Nearest neighbors

frog

frogs

toad

litoria



leptodactylidae



rana



book

books

foreword

author

published

preface

asthma

bronchitis

allergy

allergies

arthritis

diabetes

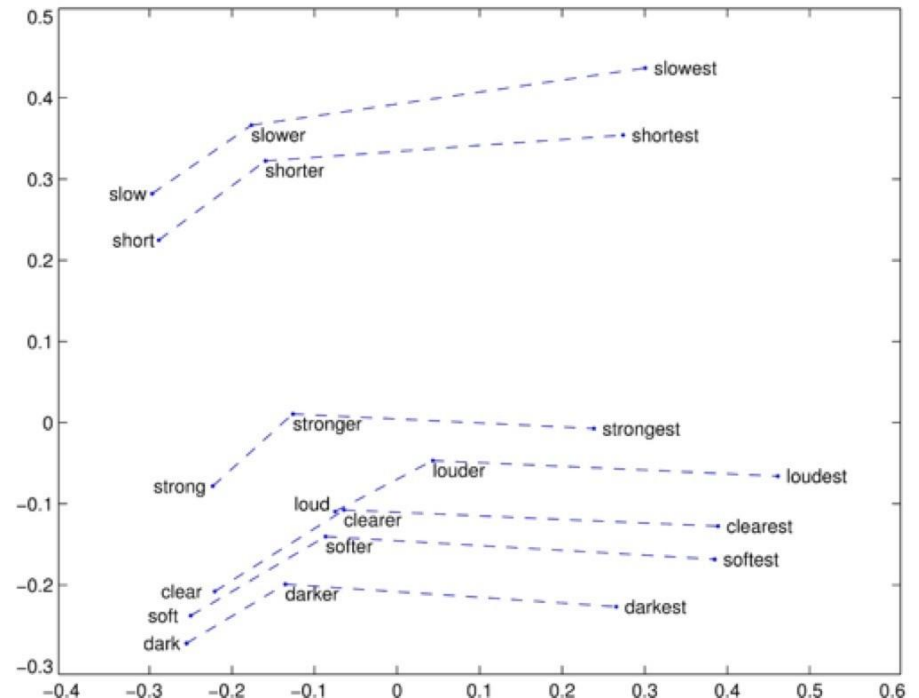
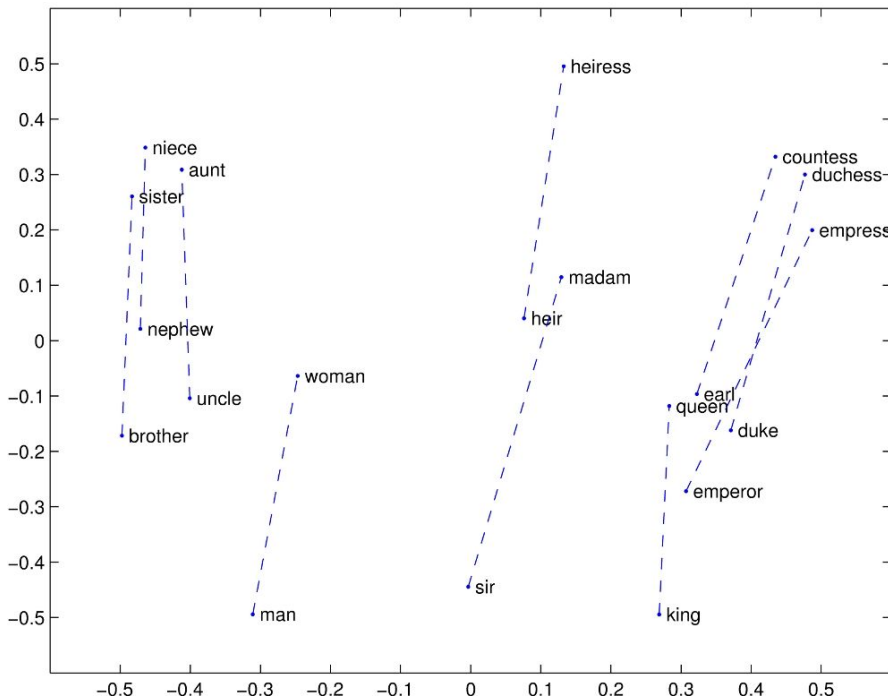
Word Embedding – Linear substructures

- Analogy task:

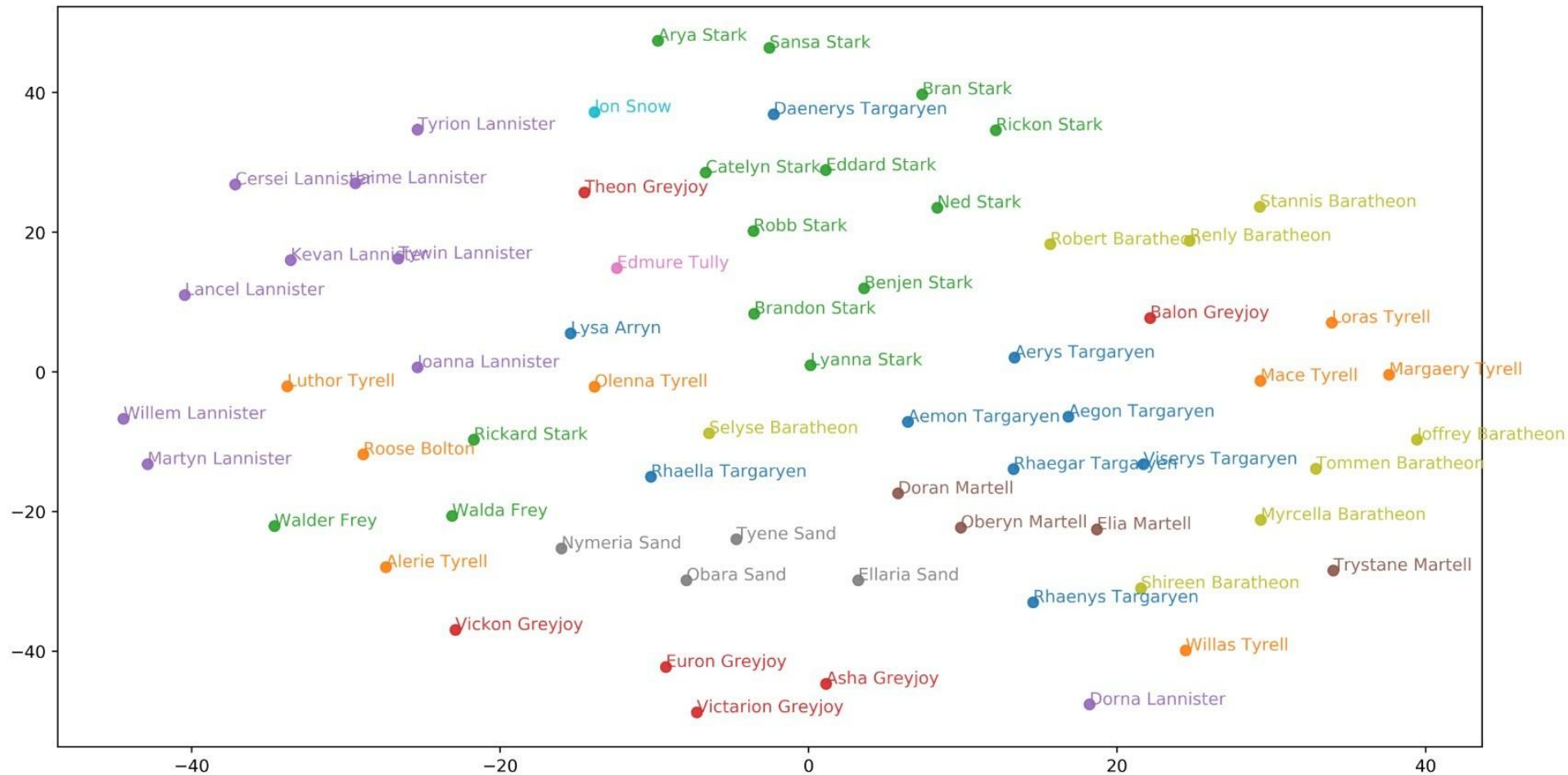
- *man to woman is like king to ? (queen)*

$$\mathbf{e}_{\text{king}} - \mathbf{e}_{\text{man}} + \mathbf{e}_{\text{woman}} = \mathbf{e}^*$$

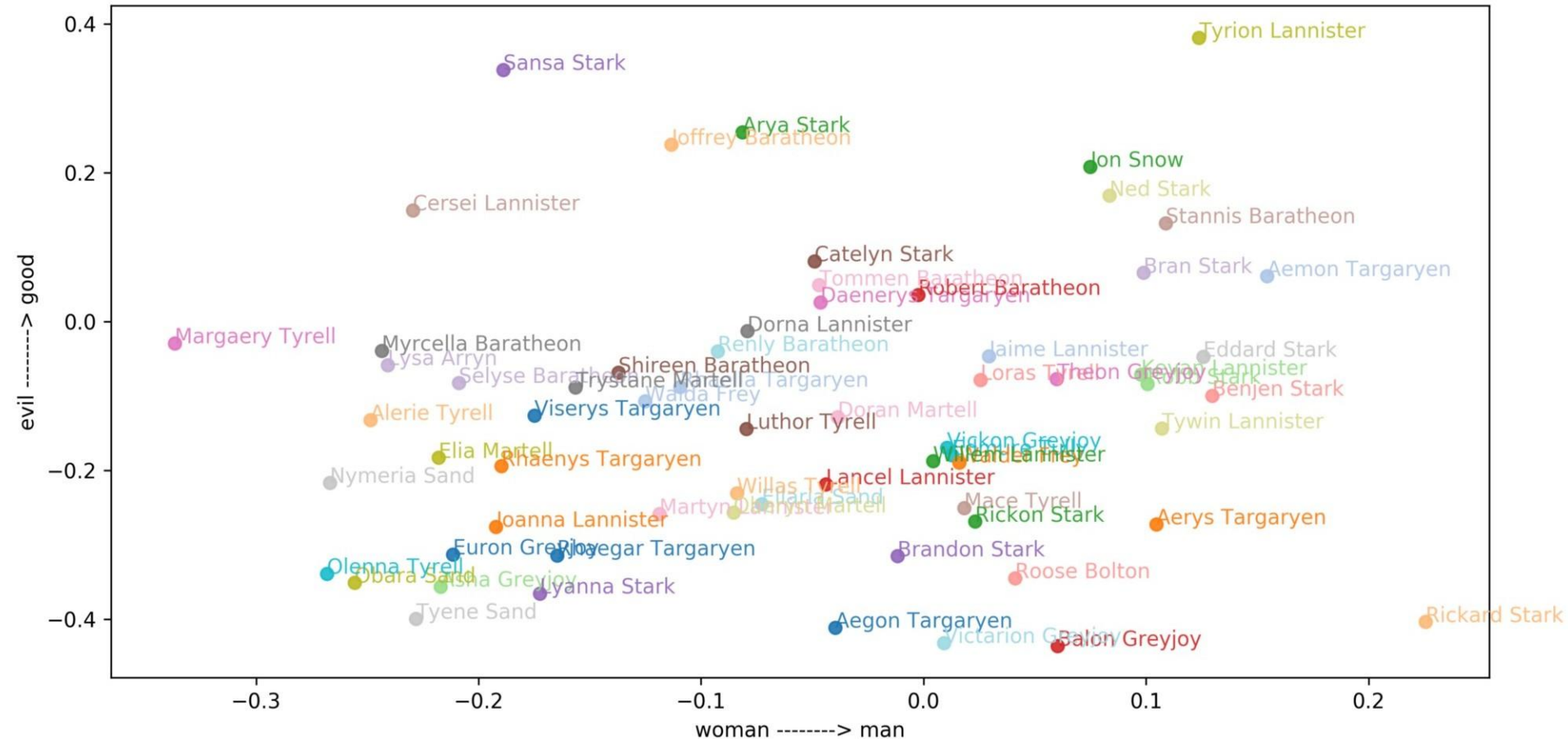
$$\mathbf{e}^* \approx \mathbf{e}_{\text{queen}}$$



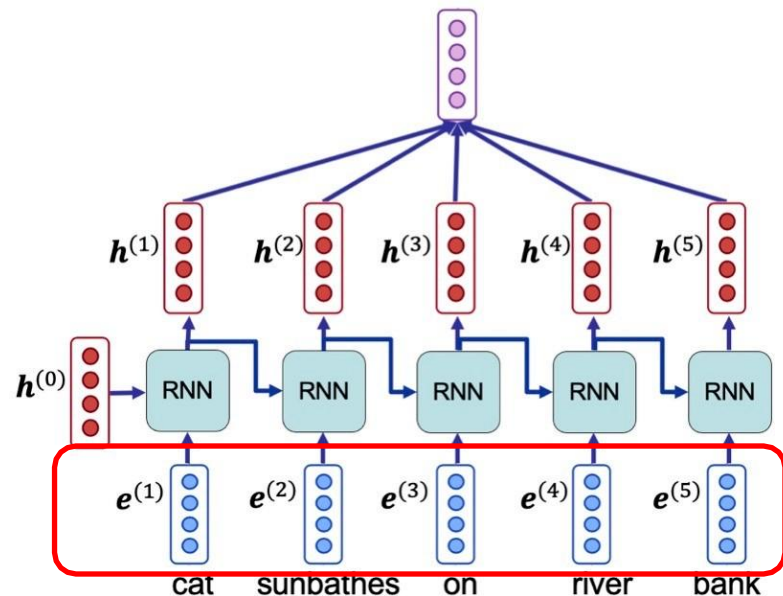
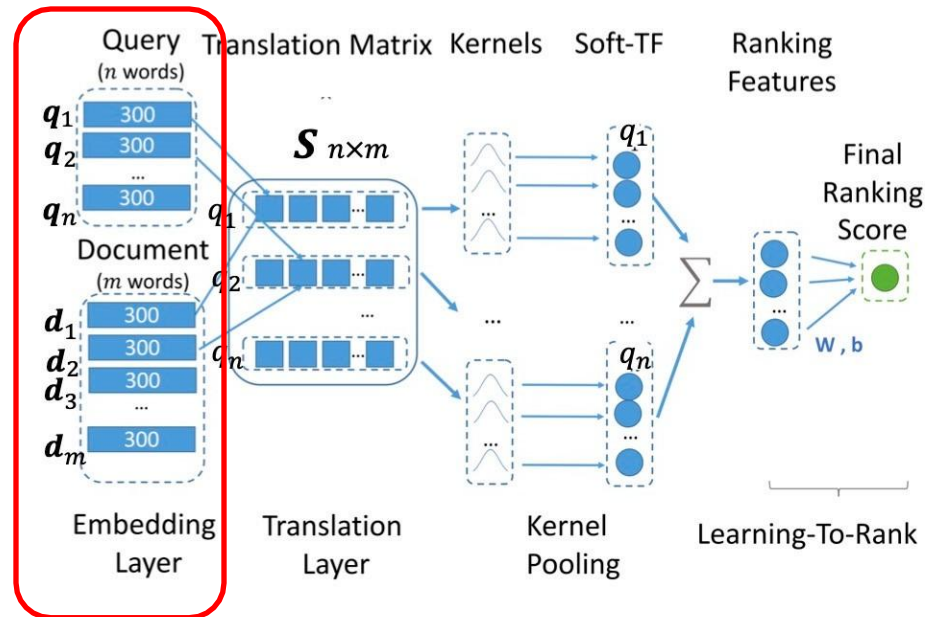
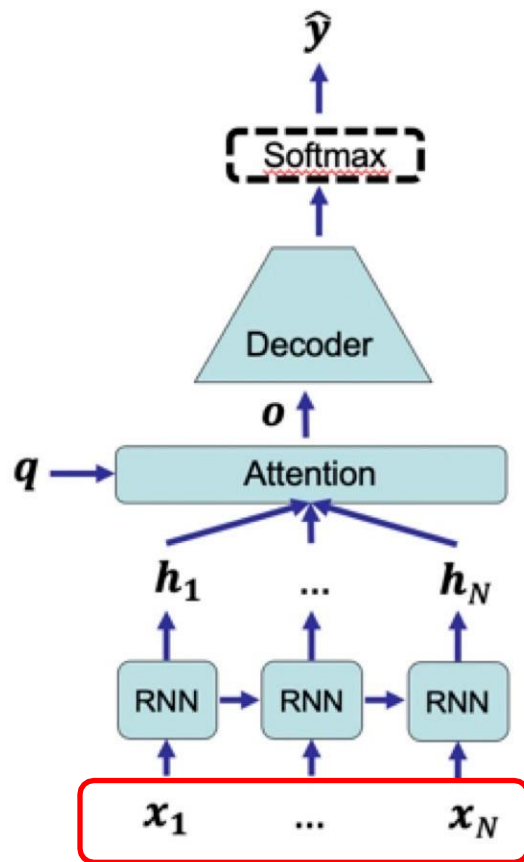
Word Embedding – Games of Thrones!



Word Embedding – Games of Thrones!



Word Embedding – Building Block of Modern NLP



Agenda

- Distributional semantics & word embedding
- **Point Mutual Information**
- GloVe

Word-Document Matrix – recap

- $\mathbb{D}$ is a set of documents (plays of Shakespeare)
 $\mathbb{D} = [d1, d2, \dots, dM]$
- $\mathbb{V}$ is the set of words (vocabularies) in dictionary
 $\mathbb{V} = [v1, v2, \dots, vN]$
- Words as rows and documents as columns
- Values: number of occurrences of words in documents
- Matrix size $N \times M$

	<i>d1</i>	<i>d2</i>	<i>d3</i>	<i>d4</i>
	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	1	8	15
soldier	2	2	12	36
fool	37	58	1	5
clown	6	117	0	0
...	...	...	...	...

Cosine

- Cosine is the normalized dot product of two vectors
 - between -1 and +1

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \cdot \frac{\mathbf{y}^T}{\|\mathbf{y}\|_2} = \frac{\sum_i x_i y_i}{\sqrt{\sum_i x_i^2} \sqrt{\sum_i y_i^2}}$$

- $\mathbf{x} = [1 \quad 1 \quad 0] \quad \mathbf{y} = [4 \quad 5 \quad 6]$

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{1 * 4 + 1 * 5 + 0 * 6}{\sqrt{1^2 + 1^2 + 0^2} \sqrt{4^2 + 5^2 + 6^2}} = \frac{9}{\sim 12.4}$$

Word-Document Matrix

	<i>d1</i>	<i>d2</i>	<i>d3</i>	<i>d4</i>
	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	1	8	15
soldier	2	2	12	36
fool	37	58	1	5
clown	6	117	0	0
...	...	...	...	...

- **Similarity** between two words:

$$\text{similarity}(\text{soldier}, \text{clown}) = \cos(\mathbf{e}_{\text{soldier}}, \mathbf{e}_{\text{clown}})$$

Context

- Context can be
 - Document
 - Paragraph, tweet
 - **Window of (usually 2 to 10) context words on each side of the word**
- Word-Context matrix
 - Every context-word as a dimension
$$\mathbb{C} = [c1, c2, \dots, cL]$$
 - Usually, $\mathbb{C} = \mathbb{V}$ and therefore $L = N$
 - Word-Context matrix is in size $N \times L$ or mostly $N \times N$

Word-Context Matrix

- Window context of 7 words
 - Values are the number of co-occurrences of v and $c \rightarrow x_{v,c}$

sugar, a sliced lemon, a tablespoonful of their enjoyment. Cautiously she sampled her first well suited to programming on the digital for the purpose of gathering data and

apricot preserve or jam, a pinch each of,
pineapple and another fruit whose taste she likened
computer. In finding the optimal R-stage policy from
information necessary for the study authorized in the

	$c1$ aardvark	$c2$ digital	$c3$ data	$c4$ fruit	$c5$ result	$c6$ sugar
$v1$ apricot	0	0	1	3	0	2
$v2$ pineapple	0	0	0	1	0	1
$v3$ computer	0	2	4	0	1	0
$v4$ information	0	4	3	0	2	0

Word-to-Word Relations

	<i>c1</i>	<i>c2</i>	<i>c3</i>	<i>c4</i>	<i>c5</i>	<i>c6</i>
	aardvark	digital	data	fruit	result	sugar
<i>v1</i> apricot	0	0	1	3	0	2
<i>v2</i> pineapple	0	0	0	1	0	1
<i>v3</i> computer	0	2	4	0	1	0
<i>v4</i> information	0	4	3	0	2	0

- First-order co-occurrence relation
 - Words that appear close-by in contexts
 - Each cell $x_{v,c}$ of word-context matrix (**computer** to **data** in the example)
 - Examples of strong first-order relations: **drink** to **rosé** or **drink** to **wine**
- High-order co-occurrence relation
 - Words that share similar contexts
 - Reflects the semantic similarity between words, and commonly measured by cosine (**computer** to **information** in the example)
 - Examples of strong high-order relations: **rosé** to **wine**, **Tesgüino** to **Märzen**, or **frog** to **toad**

Point Mutual Information (PMI)

- Problem with using raw co-occurrence statistics ($x_{v,c}$) in word-context matrix
 - Highly frequent words (“**and**”, “**the**”) co-occur with many words and gain high values, although they don’t convey much information
- Point Mutual Information (PMI)
 - Rooted in information theory
 - A common measure of **first-order co-occurrence** used to create word-context matrix
 - Defined as the joint probability of two events (random variables) divided by their marginal probabilities

$$\text{PMI}(A, B) = \log_2 \frac{P(A, B)}{P(A)P(B)}$$

Point Mutual Information (PMI)

- PMI of the occurrences of word v and context-word c

$$\text{PMI}(v, c) = \log_2 \frac{P(v, c)}{P(v)P(c)}$$

$P(v, c)$: probability of co-occurrence of v with c

$$P(v, c) = \frac{x_{v,c}}{|\mathcal{D}|}$$

$x_{v,c}$: number of times that v and c appear in the same context

$|\mathcal{D}|$: number of all co-occurrences in the corpus $\rightarrow |\mathcal{D}| = \sum_{i=1}^N \sum_{j=1}^L x_{i,j}$

$P(v)$: probability of co-occurrence of v with any context-word

$$P(v) = \frac{\sum_{\check{c} \in \mathbb{C}} x_{v,\check{c}}}{|\mathcal{D}|} = \frac{X_v}{|\mathcal{D}|}$$

$P(c)$: probability of co-occurrence of c with any word

$$P(c) = \frac{\sum_{\check{v} \in \mathbb{V}} x_{\check{v},c}}{|\mathcal{D}|} = \frac{X_c}{|\mathcal{D}|}$$

Point Mutual Information (PMI)

Word-context matrix with raw co-occurrences ($x_{v,c}$)

	$c1$	$c2$	$c3$	$c4$	$c5$	$c6$
	aardvark	digital	data	fruit	result	sugar
$v1$ apricot	0	0	1	3	0	2
$v2$ pineapple	0	0	0	2	0	1
$v3$ computer	0	2	4	0	1	0
$v4$ information	0	4	3	0	2	0

PMI ($v = \text{information}, c = \text{data}$) = ?

$$P(v = \text{information}, c = \text{data}) = 3 / 25 = 0.12$$

$$P(v = \text{information}) = 9 / 25 = 0.36$$

$$P(c = \text{data}) = 8 / 25 = 0.32$$

$$\text{PMI}(v = \text{information}, c = \text{data}) = \log_2 \frac{0.12}{0.36 \times 0.32} = 0.058$$

Point Mutual Information (PMI)

Word-context matrix with raw co-occurrences ($x_{v,c}$)

	$c1$	$c2$	$c3$	$c4$	$c5$	$c6$
	aardvark	digital	data	fruit	result	sugar
$v1$ apricot	0	0	1	3	0	2
$v2$ pineapple	0	0	0	2	0	1
$v3$ computer	0	2	4	0	1	0
$v4$ information	0	4	3	0	2	0

Word-context matrix with PMI

	$c1$	$c2$	$c3$	$c4$	$c5$	$c6$
	aardvark	digital	data	fruit	result	sugar
$v1$ apricot	$-\infty$	$-\infty$	-0.94	1.32	$-\infty$	1.47
$v2$ pineapple	$-\infty$	$-\infty$	$-\infty$	1.73	$-\infty$	1.47
$v3$ computer	$-\infty$	0.25	0.83	$-\infty$	0.25	$-\infty$
$v4$ information	$-\infty$	0.88	0.05	$-\infty$	0.88	$-\infty$

Positive Point Mutual Information (PPMI)

- PPMI sets negative values of PMI to zero

$$\text{PPMI}(t, c) = \max(\text{PMI}, 0)$$

Word-context matrix with PMI

	<i>c1</i>	<i>c2</i>	<i>c3</i>	<i>c4</i>	<i>c5</i>	<i>c6</i>
	aardvark	digital	data	fruit	result	sugar
<i>v1</i> apricot	$-\infty$	$-\infty$	-0.94	1.32	$-\infty$	1.47
<i>v2</i> pineapple	$-\infty$	$-\infty$	$-\infty$	1.73	$-\infty$	1.47
<i>v3</i> computer	$-\infty$	0.25	0.83	$-\infty$	0.25	$-\infty$
<i>v4</i> information	$-\infty$	0.88	0.05	$-\infty$	0.88	$-\infty$

Word-context matrix with PPMI

	<i>c1</i>	<i>c2</i>	<i>c3</i>	<i>c4</i>	<i>c5</i>	<i>c6</i>
	aardvark	digital	data	fruit	result	sugar
<i>v1</i> apricot	0	0	0	1.32	0	1.47
<i>v2</i> pineapple	0	0	0	1.73	0	1.47
<i>v3</i> computer	0	0.25	0.83	0	0.25	0
<i>v4</i> information	0	0.88	0.05	0	0.88	0

Summary – PPMI

- PPMI captures first-order co-occurrences
- PPMI word-context matrix provides high-dimensional word representations. The matrix is ...
 - highly sparse
 - very big, and barely fits to memory. It usually requires sparse data structures
- As shown empirically, PMI overly favors low-frequency/rare words

Agenda

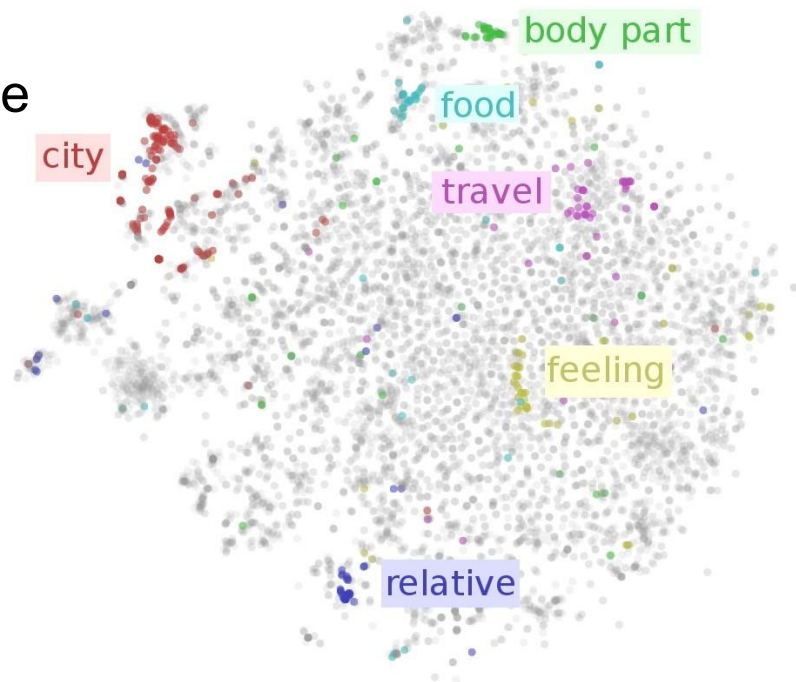
- Distributional semantics & word embedding
- PMI+SVD
- **GloVe**

Resource used in some slides:

<http://web.stanford.edu/class/cs224n/slides/cs224n-2020-lecture02-wordvecs2.pdf>

GloVe: Global Vectors for Word Representations

- A well-known word embedding model
 - Pre-trained word embeddings are typically used in various tasks
- Similar to PPMI+SVD, GloVe ...
 - starts from a high-dimensional sparse word-context co-occurrence matrix
 - and then applies matrix factorization to build low-dimensional word embeddings



Pennington, Jeffrey, Richard Socher, and Christopher D. Manning. "Glove: Global vectors for word representation." *Proceedings of the conference on empirical methods in natural language processing (EMNLP)*. 2014.

Picture: <https://ruder.io/word-embeddings-1/>

Word-Context matrix – from PMI to GloVe

- Deriving GloVe from PMI ...
- Recall:

$$\begin{aligned}\text{PMI}(v, c) &= \log_2 \frac{P(v, c)}{P(v)P(c)} \approx \log \frac{P(v, c)}{P(v)P(c)} \\ &= \log \frac{x_{v,c} / |\mathcal{D}|}{X_v / |\mathcal{D}| \cdot X_c / |\mathcal{D}|} = \log \frac{x_{v,c} |\mathcal{D}|}{X_v X_c} \\ &= \log x_{v,c} - \log X_v - \log X_c + \log |\mathcal{D}|\end{aligned}$$

- $\log |\mathcal{D}|$ is a constant and can be removed:

$$\approx \log x_{v,c} - \log X_v - \log X_c$$

Word-Context matrix – from PMI to GloVe

- Deriving GloVe from PMI ...

$$\text{PMI}(v, c) \approx \log x_{v,c} - \log X_v - \log X_c$$

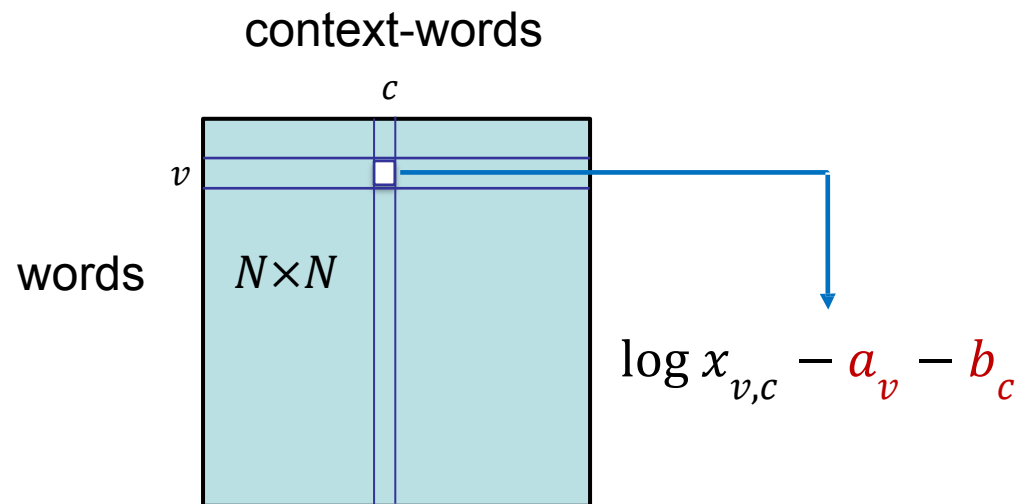
- 1st difference:** replace $\log X_v$ and $\log X_c$ with learning parameters a_v and b_c , respectively
- GloVe therefore uses:

$$\log x_{v,c} - a_v - b_c$$

- a_v and b_c are **bias terms**
- In fact, a_v and b_c act as a “**normalization**” to log-co-occurrence, where their degrees are learned
- Vector **a** contain the bias values of all words (like a_v)
- Vector **b** contain the bias values of all context-words (like b_c)

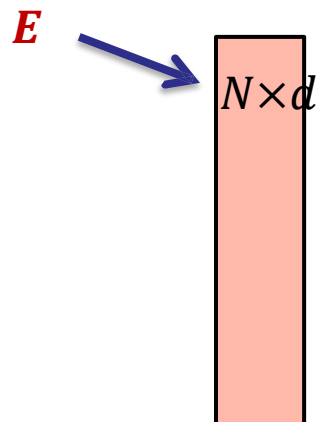
Matrix factorization

GloVe word-context matrix:

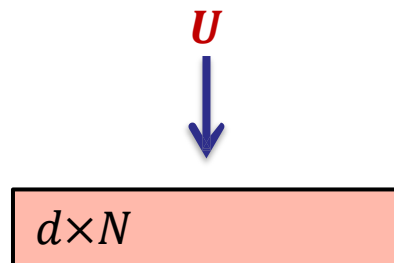


2nd difference: matrix factorization

word vectors

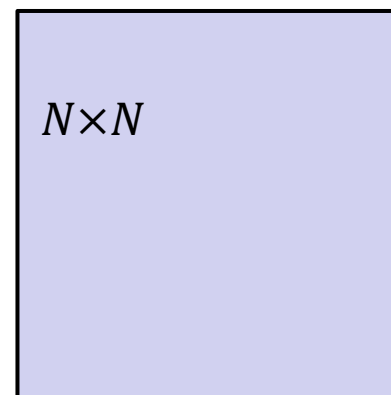


context-word vectors



.

=



approximated
co-occurrence
matrix

$\updownarrow f$

Loss: make the differences
as small as possible

Matrix factorization

Formal definition:

We factorize GloVe co-occurrence matrix by defining two learnable parameter matrices of size $N \times d$:

- ❑ $\mathbf{E}$ (word embeddings)
- ❑ $\mathbf{U}$ (context-word embeddings)

such that for every (v, c) pair:

$$\mathbf{e}_v \mathbf{u}_c^T \approx \log x_{v,c} - \mathbf{a}_v - \mathbf{b}_c$$

$\mathbf{E}$, $\mathbf{U}$, $\mathbf{a}$, and $\mathbf{b}$ are (learnable) model parameters

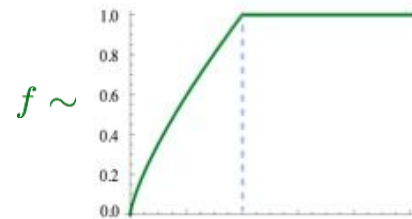
Loss function

- To find optimum values for parameters, GloVe defines the following **least squares** loss function:*

$$\mathcal{L} = \sum_{v \in \mathbb{V}, c \in \mathbb{C}} (\mathbf{e}_v \mathbf{u}_c^T + a_v + b_c - \log x_{v,c})^2$$

* GloVe includes a function f to the loss, which weighs down the rare pairs and saturates on highly frequent ones. The loss function in GloVe is **weighted least squares** (details in paper):

$$\mathcal{L} = \sum_{v \in \mathbb{V}, c \in \mathbb{C}} f(x_{v,c}) (\mathbf{e}_v \mathbf{u}_c^T + a_v + b_c - \log x_{v,c})^2$$

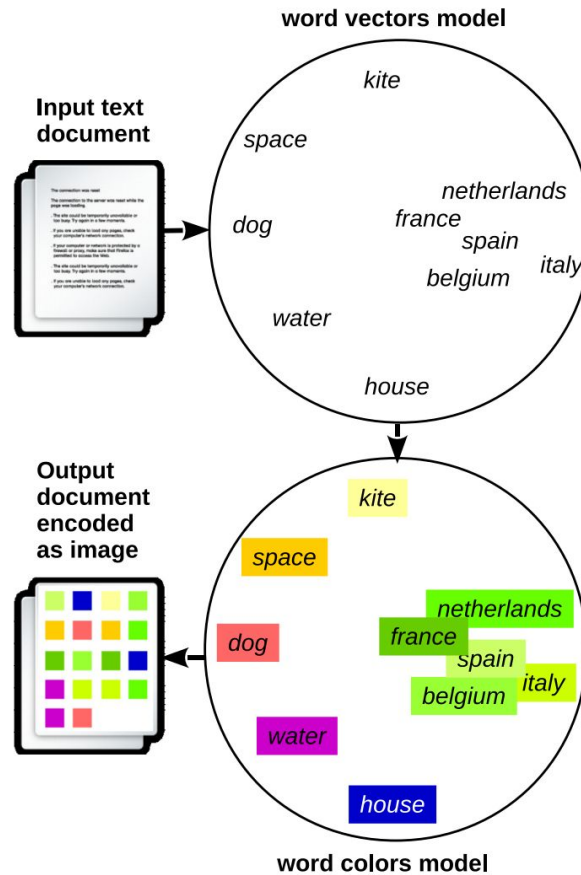


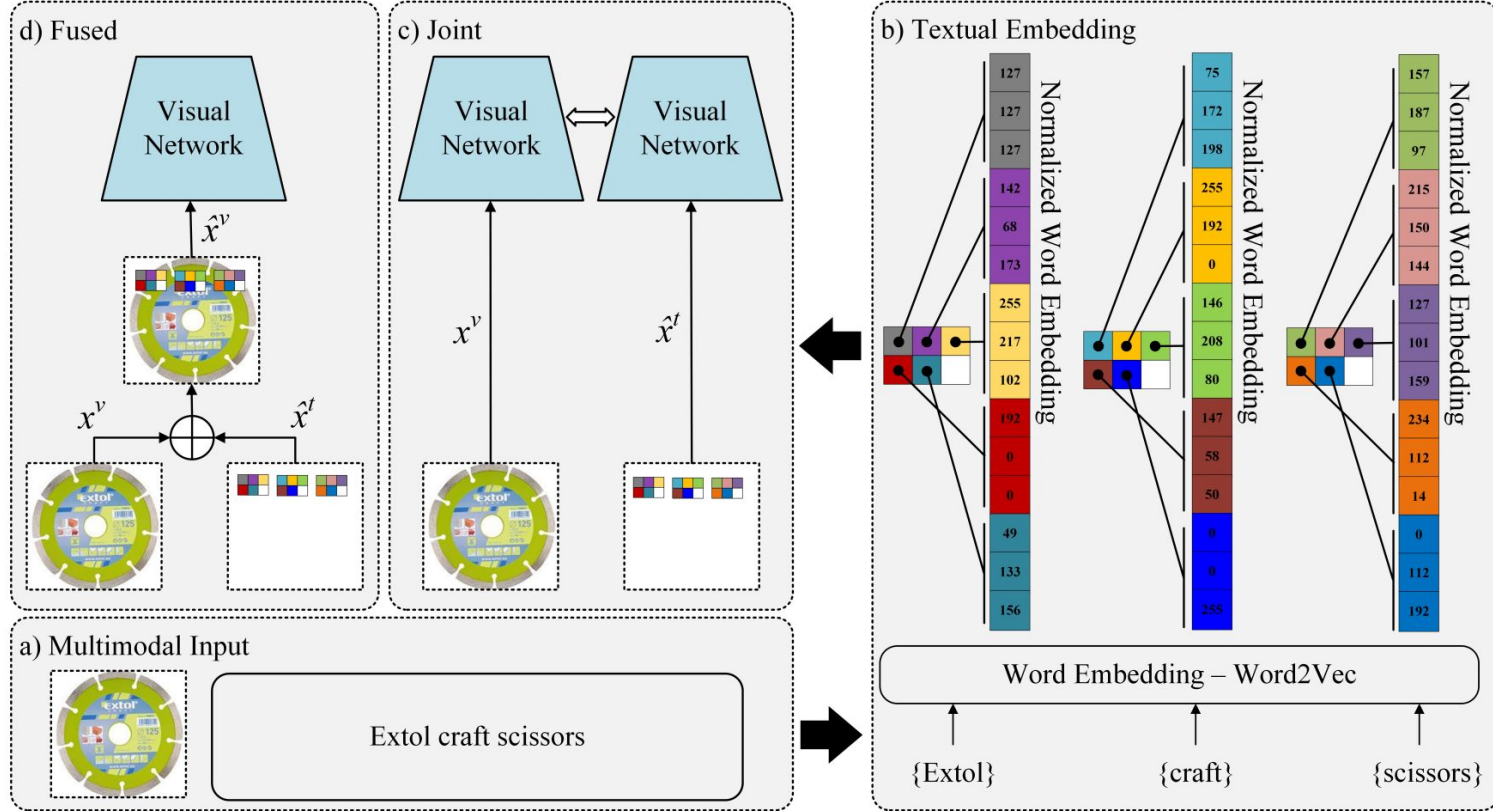
Optimization

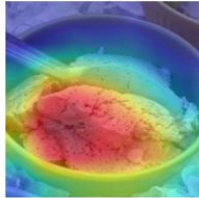
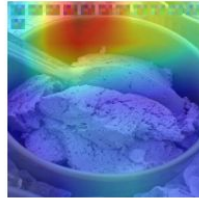
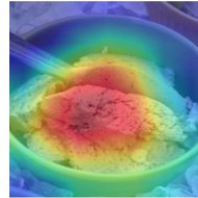
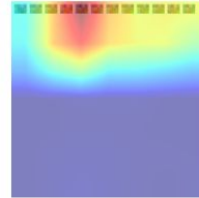
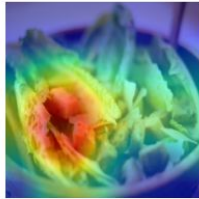
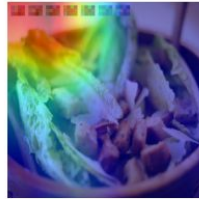
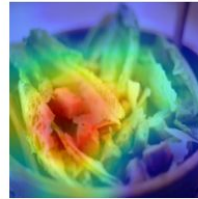
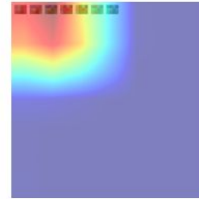
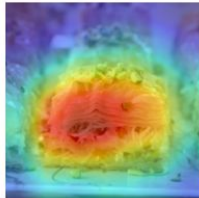
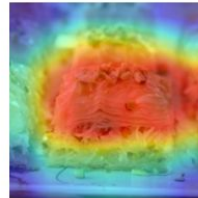
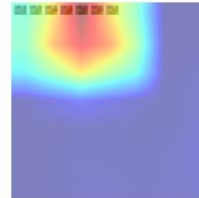
- To optimize the parameters according to the loss function, either of the following optimization methods can be used:
- Gradient descent-based methods
 - More in the next lecture
- Alternating Least Squares (ALS)
 - More about ALS is available here:
<http://stanford.edu/~rezab/classes/cme323/S15/notes/lec14.pdf>

GloVe – Summary

- GloVe
 - first accumulates co-occurrence numbers
 - then defines two embedding matrices and two bias vectors, and optimize their values by approximating the logarithm of co-occurrence statistics
- Final word embeddings
 - can be the word vectors E
 - but can also be the sum of the word and context-word vectors: $E + U^T$
 - This method is used in GloVe
- Characteristics
 - Fast training → repeat SGD till loss converges
 - Scalable to large corpora
 - Good performance even with small corpus and small vectors





		Baseline		Chameleon		
		a. Orig. Image	b. Unimodal	c. Multimodal	d. Missing Text	e. Missing Image
Ice Cream						
		0.60	0.99	0.99	1.0	
Caser Salad						
		0.98	0.99	0.99	1.0	
Baklava						
		0.98	0.99	0.99	0.99	